

Visualizing Player Movement and Game Events through Space-Time Cubes in Extended Reality

Niko S. Nannemann*

Shivam Agarwal†

Fabian Beck‡

University of Bamberg

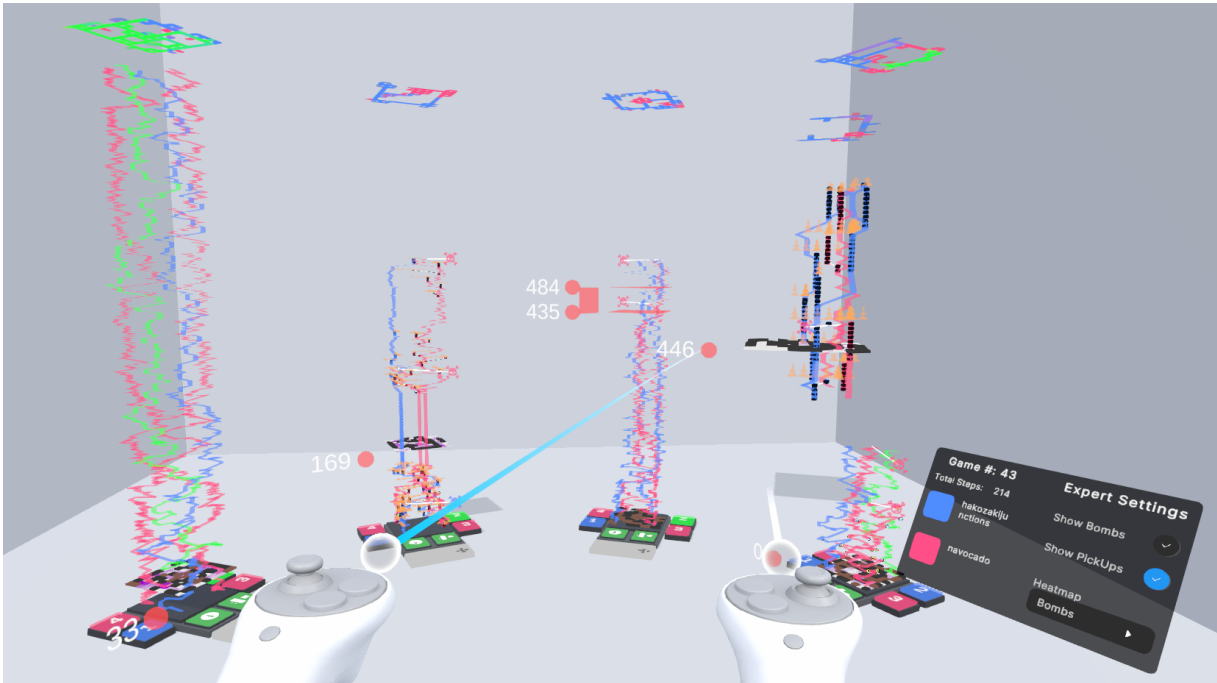


Figure 1: Four games displayed as space-time cubes. From left to right with different interactive selections: Game #15 with Agent 2 highlighted (green trajectory) and the game board at Step 33; Game #28 with bombs visible and the game board at Step 169; Game #6 with time range selection enabled (Step 435 to Step 484); Game #43 with active info panel and item pickups.

ABSTRACT

To understand multiplayer gameplay behavior in a bomb-laying game, this paper explores the visual representation of player movement and events using space-time cubes. Complementing a previous two-dimensional event visualization, the approach focuses on contextualizing the player trajectories on the board with important events. Leveraging extended reality technology, the three-dimensional space-time-cube representation of a game session can be placed like columns in the virtual space. Various techniques support the interactive exploration of the spatiotemporal data. We demonstrate insights that can be found through the analysis of AI agent play behavior.

Index Terms: Game analytics, space-time cube, event visualization, immersive analytics, extended reality.

*e-mail: niko-sebastian.nannemann@stud.uni-bamberg.de

†e-mail: shivamworking@gmail.com

‡e-mail: fabian.beck@uni-bamberg.de

1 INTRODUCTION

Spatiotemporal movement data, especially when enriched with further information such as events, is challenging to visualize. Ideally, we want to see *when* and *where* something happens, and compare these situations across time. When the spatial data is two-dimensional, we can use a third dimension for time in a *space-time cube* [13], creating an artificial three-dimensional representation that makes temporal changes spatially explorable. However, three-dimensionality is not without challenges. Particularly concerning occlusion, depth perception, and interaction, which can be problematic when presented with conventional two-dimensional displays and desktop setups [18, 8]. To address these issues, *immersive analytics* explores emerging technologies such as *virtual reality* (VR) or, more generally, *extended reality* (XR) for data analysis [11, 6]. Users can effectively deal with occlusion and visual clutter by simply changing their perspective through natural head movements [8].

Game analytics [21, 7] concerns the investigation of player behavior in games and often faces the challenge of exploring such spatiotemporal data. To improve their gameplay, players try to understand the strategies of their opponents. Moreover, games are used as testbeds for intelligent artificial agents and analyzing the game sessions helps improve the agents. In board-based games, one or multiple players (or game pieces) move across the board and perform actions that can be considered as events. We specifically study the

classic computer game of *Bombberman*, and its open variant *Pommerman* used in artificial intelligence (AI) competitions [16]. In this game played on a two-dimensional board, two teams compete on a board with destroyable and non-destroyable blocks by laying bombs and trying to eliminate each other. Previous work [2] has already shown how timeline-based event visualization can provide rich insights into player behavior. However, the spatial component was restricted to a video replay of the match.

In this work, we explore the potential of space-time-cube representations in an extended-reality setting as a complementing means to the previous two-dimensional, event-focused visualization (note that, while the analysis is here done in extended reality, the game itself is not played in this environment, but on a two-dimensional board without three-dimensional graphics). Emphasizing the spatiotemporal aspect, as shown in Fig. 1, the board forms the ground plane of the three-dimensional, column-like game session representation, where time progresses from bottom to top. The space-time cubes visualize the players—here, referred to as *agents* because of studying artificial players—and their movements as colored trajectories, assigned to the red and blue team. Various visual augmentations can be activated including bombs, their movement, and explosions, as well as power-up collection by agents and death events. Since the space-time cube showing the whole game session might be too dense for a detailed analysis, an interactive time range selection allows for exploring an enlarged temporal cut-out in a second space-time representation (Fig. 1, right). Moreover, time-aggregating, heatmap-like visualizations are available to summarize agent movement and bomb placement. The visualizations are available in a virtual-reality and a camera pass-through mode; while we prefer the camera pass-through mode when interactively using the tool for easier maneuvering in a restricted space (like an office), we mostly present the static screenshots (especially, dense overview visualizations) in this work in VR mode for a cleaner background. The supplemental material includes a video demonstration using the pass-through mode.

2 RELATED WORK

The idea of space-time cubes [13]—a visualization approach that maps 2D spatial data on the two axis and shows time on the third axis—has been influential in analyzing spatiotemporal data. They were also extended to represent event data along with spatiotemporal attributes [10, 14]. While initially mostly built for two-dimensional displays, head-mounted displays and immersive analytics presented new ways of analyzing data [19, 6, 9]. Space-time cubes have also been proposed for such environments, and found to be more preferred over desktop-based implementation by users and received higher usability scores [8]. The reasons being a natural experience with the spatial context and reduced mental workload to analyze a ‘static’ visualization vs. animation-based approaches to analyze temporal changes. Wagner et al. [20] propose a three-dimensional immersive framework using a space-time-cube metaphor with an extended query model and user interactions. As well, closely related to our work is a space-time-cube visualization for three-dimensional eye gaze data in virtual reality [15] and using situated space-time cubes for analytics [3]. However, these immersive approaches do not visualize event sequence data together with the spatial context.

Game analytics uses diverse visualizations to understand gameplay behaviors and uncover insights [21], especially for understanding player behavior in multiplayer games [7]. In game analytics, space-time-cube visualizations have been found useful as well. For instance, Coulton et al. [5] used space-time cubes for player movement paths and their analysis in location-based games. However, they do not visualize space-time cubes in an immersive environment. Such immersive space-time-cube representations have been proposed by Horst et al. [11] for *League of Legends* with cus-

tomized three-dimensional glyphs, encoding match data (e.g., gold, kills, the number of minions and champions killed) in the position and properties of the glyph. This helps in tracking spatiotemporal gameplay patterns (map coverage) and unusual trajectory of players. However, event related insights are difficult to spot.

3 BACKGROUND AND DESIGN GOALS

Our approach analyzes data from a specific game and complements a previous analysis approach. We first introduce the *Pommerman* game environment, as well as the *Bombalytics* visualization approach and propose extended design goals that guided developing our approach.

The *Pommerman* Game. *Pommerman* [16] is a multi-agent bomb laying game inspired by the classic *Bombberman* game. It has been developed as an open testbed by AI researchers to explore AI techniques for building intelligent agents (e.g., [12]). Two teams of each two agents compete to eliminate the opposing agents. Each game takes place on an 11×11 grid with symmetric agent placement, navigable spaces, rigid walls, and destroyable wooden walls hiding power-ups. Agents eliminate opponents by placing bombs that detonate after ten steps, producing flames that persist for three turns. On each turn, agents can move, lay a bomb, or idle. Games end in a draw if the step limit is reached.

***Bombalytics* Visualization Approach.** To analyze strategies exhibited by the AI agents in *Pommerman*, Agarwal et al. [2] introduced *Bombalytics*, a horizontal timeline-based visualization. Agent actions (e.g., laying a bomb), game events (e.g., bomb explosion), and interactions (e.g., picking power-ups, kicking bombs) are mapped to rows—agents from teams A and B at the top and bottom, and shared resources (power-ups) in the center. The interface also includes a game playback for spatial context but no further encoding of spatial information in the timeline, which was clearly criticized by users in an evaluation. *Bombalytics* was built using three design goals: To show an overview of events in a game (G1), map local patterns and repetitions to analyze exhibited strategies (G2), and provide an overview of multiple games (G3).

Extended Design Goals. To strengthen the spatial aspect of the analysis, we extend the design goals for this work:

G4: Integrated Spatiotemporal Analysis. Always within their spatial context (i.e., the board), the approach should support in an integrated analysis of (a) agent movement, (b) bombs and their explosions, as well as (c) other events.

G5: Overview of Spatial Patterns. The approach should aggregate the data across time to provide an overview of movement and spatial event patterns.

While the previous design goals stay valid as secondary criteria, the additional ones are the focus and priority of the new approach presented in the following, which is intended to complement the previous work, but not to replace it (see also Sec. 6).

4 VISUALIZATION APPROACH

Based on the goals, we designed our visualization approach and implemented it in a prototype named *PomXR*. It was developed using *Unity* with the *OpenXR* library and ran on the *Meta Quest 3*.

Space-Time Cube. As a frame for an integrated spatiotemporal representation (G4), *PomXR* depicts a game session in a space-time cube that always shows the full session length (*primary STC*; Fig. 2C). As illustrated, for instance, in Fig. 2, it represents the board of the game as the two-dimensional ground plane (Fig. 2A), while showing time from the bottom to the top. The ground plane has attached buttons for grabbing and moving the space-time cube (⊕), toggling the information panel (Ⓜ), enabling

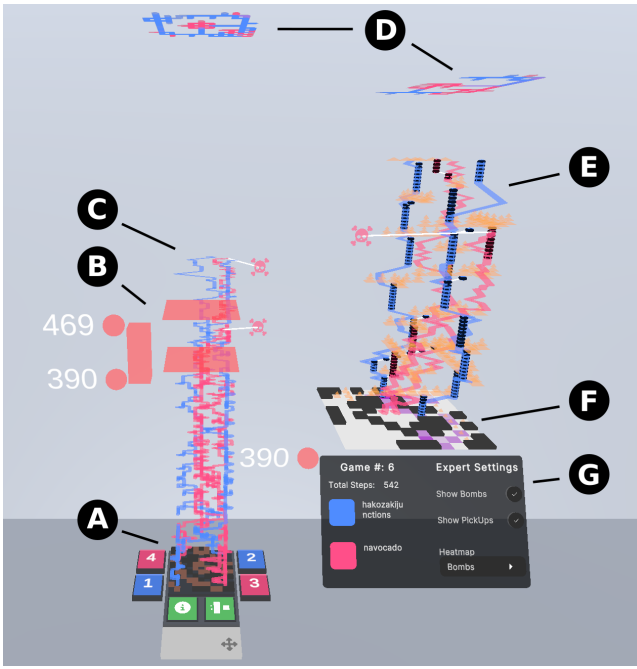


Figure 2: Annotated visualization of Game #6 with an active range selection: (A) ground plane, (B) range selector, (C) primary STC, (D) time-flattening view, (E) secondary STC for the selected range, (F) game board, (G) info panel. Death events are visible in both STCs, the bombs and flames only in the secondary STC.

the time-range selection (B), and numbered buttons allowing users to highlight the respective agent. For comparing multiple games (G3), we can juxtapose the respective space-time cubes using the same height per timesteps, as shown in Fig. 1.

Agent Movement. We consistently use colors to discern the agent’s movements by team (red and blue). Hence, the movement of agents is visualized as red or blue trajectories within the *primary STC* (Fig. 2C). To mitigate occlusion, they are drawn as semi-transparent shapes. Just visualizing the trajectories like this is already sufficient to get insights into potential teamwork by the agents and to see what region is dominated by which team (G4a). In addition, to give an overview of the agent movements (G5), we incorporate an aggregation at the top (Fig. 2D), proposed as the *time flattening* operation in space-time cubes by Bach et al. [4]. As an orthographic projection of the semi-transparent trajectories to the top plane, they form a kind of *heatmap* of player movement, blending red and blue colors when they visited the same fields.

Bombs. We enrich this representation with information about bomb laying and explosions (G4b). Bombs are depicted as small gray spheres with colored tops and bottoms to identify the agent’s team (Fig. 2E). Since bombs can be kicked, the spheres at different timesteps are connected by a white line to trace their trajectories. Bomb explosions are visualized as orange, semi-transparent pyramids, resembling an abstract flame. As the flame’s life decreases, its base shrinks each time step, signaling reduced danger and a sooner escape for trapped agents. This encoding implicitly visualizes, as a temporal sequence, bomb laying and bomb kicking events, as well as explosions (G1). In addition, for the analysis of areas with the highest density of laid bombs (G5), a heatmap of bomb positions aggregated over time is shown with darker purple shades on the game board (Fig. 2F).



Figure 3: In Game #28, the blue agent traps itself placing bombs at both entrances to its area and is destroyed by its own bomb. Meanwhile, the other agents, farther away, focus on collecting power-ups.

Events. Further events are mapped directly within the space-time cube (G1, G4c). Identifying the collection of different power-ups using icons (see Fig. 3), an agent might acquire the ability to lay extra bombs (at the same time) (G), to extend bomb flame ranges (G), and to kick a bomb (G). Since an agent’s death is a prominent event, we show this using an event marker (G, [1]), circling the space-time cube, which is connected to the spatial origin via a white leader line (e.g., Fig. 2, Fig. 4). The icons always face the user to enhance visibility and reduce occlusion.

Interactive Exploration. To provide the exact local spatial context of an agent’s movement or events (G2, G4), the game board can be moved along the temporal axis in the space-time cube (e.g., Fig. 1, Fig. 4). To facilitate a more in-depth analysis of a specific section of the game (e.g., with substantial movements, or deaths), we integrate a *time chopping* interaction [4] by allowing users to select a specific time range (Fig. 2B), which can be resized or moved along the temporal axis. The extracted section is then shown as a secondary juxtaposed enlarged space-time cube (*secondary STC*, Fig. 2E). With this interplay of the two space-time cubes, the *primary STC* effectively acts as a visually enriched scrollbar for the zoomed-in *secondary STC*.

5 APPLICATION EXAMPLE

We used the proposed visualization approach to analyze games between the top teams (*hakoakikujunctions* in blue and *navocado* in red) of the *NeurIPS 2018 Pommerman Competition* [17]. We selected four games for investigation (from left to right in Fig. 1: #15, #28, #6, and #43), aligning with those analyzed by Agarwal et al. [2] using *Bombalytics*. However, in the following, due to space restrictions, we only highlight the most important findings of two games (#28, #6).

Early Death. In Game #28, we observed an early defeat of one of the blue agents (G1). A closer inspection reveals that the agent had placed bombs at the two entrances of its area (G4b). Although it had a chance to avoid the flames, the agent stopped moving and was ultimately destroyed by its own bomb (G4a; Fig. 3). This clearly hints at optimization potential for the blue team.

Oscillation. In the same game, we observed that the other blue agent oscillated between two positions for several time steps later in that game (G4; Fig. 1, second left), indicating confusion in decision-making. While this insight was also observed in *Bombalytics* [2], it relied on the playback for confirmation. Looking

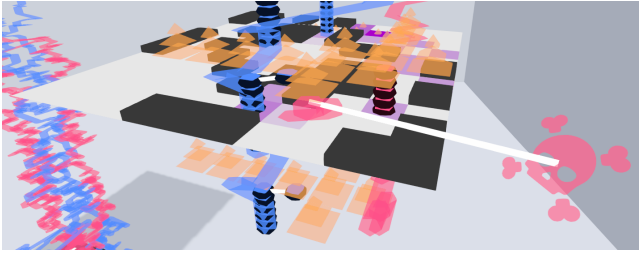


Figure 4: Close-up of the red agent’s elimination in Game #6: The red agent trapped in a confined space with both entrances blocked, one by the blue agent’s bomb, the other by its own, leaving the red agent cornered. The blue agent kicks its bomb toward the red agent.

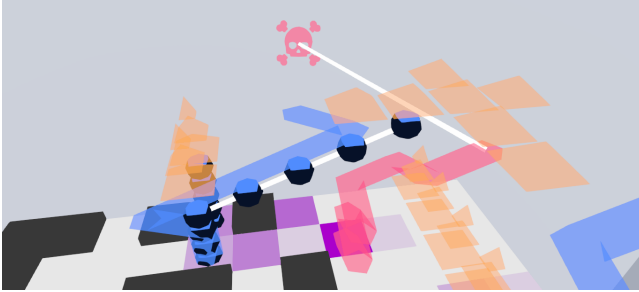


Figure 5: In Game #6, the blue agent places a bomb, waits four steps, then kicks it toward the red agent. The bomb travels five tiles and explodes on the sixth, eliminating the red agent.

closely, we selected the range (100 timesteps) and identified that the red agents did not react to this situation, stood still (vertical red lines), and did not drop any bombs.

Cornering. Game #6 featured a situation where spatial context enhanced our understanding of the elimination of a red agent. Here, the red agent found itself cornered, not only by the opposing team but also due to its own laid bomb (G4; Fig. 4). While the timeline visualization in *Bombalytics* indicated that both red and blue agents had placed bombs, and that the blue agent kicked its own bomb—resulting in the red agent’s elimination—the spatial dynamics of this encounter were not immediately obvious. Now, we see that the red agent’s defeat was not solely due to the blue agent’s strategy, but also that the red agent had trapped itself (G4).

Precise Bomb Kick. In the same game, we identified a previously undiscovered behavior of the blue agents during the analysis of the elimination of a red agent. This particular behavior involved not only waiting on laid bombs [2], but doing so with precision (G2 and G4). In one instance, the blue agent first waited on a bomb for five steps and then kicked it, causing the bomb to travel exactly four tiles before its explosion on the fifth tile, eliminating one of the red agents (Fig. 5). At times, the blue agent would kick bombs far away with the intention of detonating them in specific locations. Such insights are much more difficult to extract in *Bombalytics*, requiring switching between timeline and playback views.

Hotspots. In Game #6, activity is overwhelmingly focused in the center and one corner of the board (Fig. 2). This is evident both in the bomb-placement heatmap (G5), in the agent-trajectories (G4a), and the time flattening view (G5). A closer inspection of that board segment reveals that one (and sometimes both) red agents become confined within this cave-like enclosure in the center (Fig. 6), accessible through only two narrow entrances (G4). Blue agents capitalize on this layout by planting bombs at these choke-

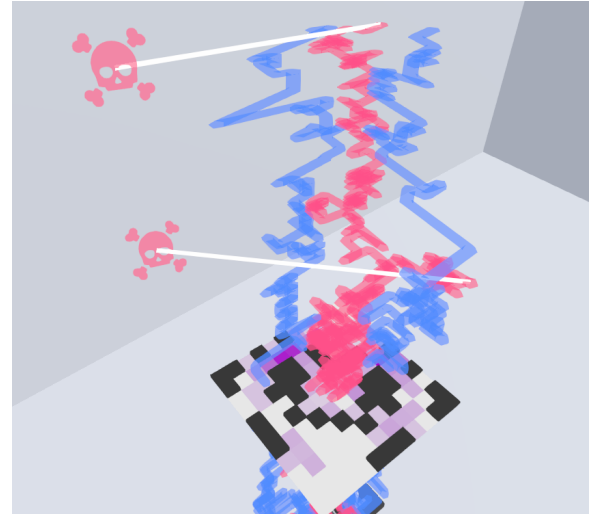


Figure 6: In Game #6, the blue agents trap the red agents in the center and one corner, by planting bombs at the chokepoints.

points, trapping the red agents (G4)—a strategy that the heatmap on the game board vividly exposes (G5).

6 DISCUSSION AND CONCLUSION

We presented a visualization approach for analyzing spatiotemporal game analytics data, including movements and events, through a space-time-cube representation in an extended reality setting. By adding emphasis on spatial aspects, such as agent movement and spatial contextualization of events, it adds to previous work [2], which highlighted temporal patterns and the event perspective. Our application examples showcase specific situations that link to the additional goals of spatial and spatiotemporal analysis. In this way, the three-dimensional approach means to complement the previous two-dimensional visualization. However, they would work best if fully integrated in the future. This can be solved simply by adding a two-dimensional virtual screen to the extended reality setting, or vice versa, embedding a three-dimensional view in the two-dimensional interface (at the cost of losing the natural navigation of the extended reality setting). Interactively linked highlighting could help to inspect the two representations together. Yet, to realize the full potential of the two approaches jointly, a more integrated design appears necessary, for instance, a two-dimensional timeline that seamlessly transforms into a space-time cube.

Our approach is tailored specifically to the game of *Pommerman*. However, the applied techniques can be transferred to related analysis scenarios, most directly the analysis of other board-based games and team sports. Depending on the specific game, challenges might arise around visualizing more trajectories and longer time spans (e.g., 22 players and one ball in soccer over 90 minutes)—efficient selection techniques would become even more important, but additional data abstractions might be necessary, too (e.g., grouping the players by role). Moreover, our approach is an example of how space-time cubes can be leveraged within an extended reality analysis setting, especially using rich interactions for exploration. This aligns with Bach et al.’s [4] proposal of various operations on general space-time cubes, highlighting such cubes as a conceptual tool for understanding temporal data. The time flattening view, along with time cutting and time chopping operations, are used already within the proposed approach. In future work, it would be relevant to investigate more systematically which of the suggested operations most suit the representation in extended reality and how they can be implemented effectively.

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